

PUBLISHED: 06-APR-2013

2015.0 RANGE ROVER (LG), 303-06

STARTING SYSTEM - TDV8 4.4L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the starting system and operation, refer to the relevant description and operation section of the workshop manual.

REFER to: [Starting System](#) (303-06E Starting System - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Starter motor ▪ Engine (turns freely)	▪ Battery ▪ Fuses ▪ Starter relay ▪ Wiring harness ▪ Damaged, loose or corroded connectors ▪ Ignition switch ▪ Generator ▪ Transmission control module ▪ Engine control module

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for diagnostic trouble codes (DTCs) and refer to the DTC index

1. Check DDW for open campaigns. Refer to the corresponding bulletins and SSM's which may be valid for the specific customer complaint and carry out the recommendations as needed.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
The engine does not crank (starter motor does not turn)	▪ Gear selector not in P or N position (vehicles with automatic transmission) ▪ Battery ▪ Starter relay ▪ Invalid key code received by central junction box ▪ Brake pedal switch failure ▪ Harness /connectors ▪ Starter motor ▪ Ignition switch ▪ Generator ▪ Transmission control module ▪ Engine control module ▪ Engine seized	Make sure the gear selector is in the P or N position and correctly adjusted. Check the battery condition and state of charge. Check for DTCs indicating an immobilizer or brake pedal switch failure. Check the starter motor relay, ignition switch and generator circuits. Refer to the electrical guides. Check for TCM and ECM DTCs. Check that the engine turns freely
The engine does not crank (starter motor does turn)	▪ Starter motor installation ▪ Starter motor ▪ Flywheel/drive plate ring gear	Check the starter motor installation (fasteners tight, starter motor square to engine, etc). Check the flywheel/drive plate ring gear teeth for damage, foreign objects, etc
Engine cranks too slowly	▪ Battery ▪ Harness /connectors ▪ Starter motor	Check the battery condition and state of charge. Check the starter motor circuits. Refer to the electrical guides. Check the engine oil grade and condition

	Oil grade	
Engine cranks too fast	Low engine compression	Check the engine compressions
Excessive starter motor noise	- Starter motor - Flywheel/drive plate ring gear - Starter motor installation /casing	Check the starter motor installation (fasteners tight, motor square to engine, etc). Check the starter motor casing condition. Check the flywheel/drive plate ring gear teeth for damage, foreign objects, etc
The engine does not crank (starter motor 'clicks')	- Low battery voltage - Battery failure - Water ingress starter motor breather tube - High resistance through connectors, terminals, megafuses, cables, and installation to the starter motor - Failure of starter motor ring gear - Failure of flywheel ring gear	GO to Pinpoint Test A .

DTC INDEX

For a list of diagnostic trouble codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4L Diesel, DTC: Module Name: Engine Control Module (100-00 General Information, Description and Operation).

PINPOINT TEST A : STARTER MOTOR PRIOR APPROVAL TEST PROCEDURE

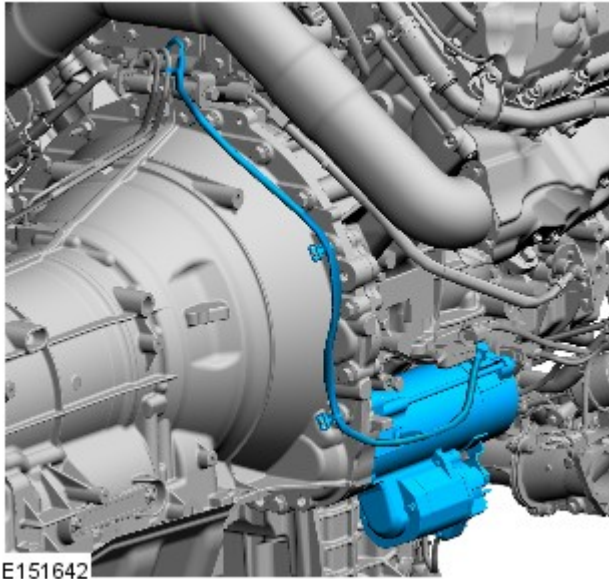
TEST CONDITIONS

DETAILS/RESULTS/ACTIONS

A1: PRE CHECKS (MANUFACTURER APPROVED BATTERY TESTER)

	1 Please refer to workshop manual section 414-01 and carry out the diagnosis and testing using the manufacturer approved test equipment. Record the information using the battery care report sheet
	Have the battery care routines using the manufacturer approved test equipment been carried out and has the information using the battery care report sheet been recorded? Yes Proceed to next step No Carry out the battery care routines using the manufacturer approved test equipment. Record the information using the battery care report sheet Proceed to next step
A2: PRE CHECKS (GENERAL WEATHER CONDITIONS)	
	 NOTE: The results of this test step may be required by dealer technical support 1 Confirm with the customer and record the weather conditions and ambient temperature (as accurately as possible) at the time of failure
	2 Confirm and record the weather conditions and ambient temperature (as accurately as possible) at the time of test
	Have the weather conditions and ambient temperature been recorded Yes Proceed to next step No Record the weather conditions and ambient temperature Proceed to next step
A3: CHECK FOR RELATED STARTER MOTOR, IMMOBILIZER DIAGNOSTIC TROUBLE CODES	
	1 Using the manufacturer approved diagnostic system, check for starter motor or immobilizer, related DTCs and refer to the relevant DTC index
	Are starter motor or immobilizer, related DTCs logged? Yes Carry out any repair action as required Clear the DTC and retest No Proceed to next step
A4: CHECK INSTALLATION OF STARTER MOTOR BREATHER TUBE	
	1 Please refer to workshop manual and check the starter motor breather tube is installed and routed correctly. The shepherds crook

needs to be positioned correctly to prevent water ingress to the starter motor



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Is the starter motor breather tube installed and routed correctly?

Yes

Proceed to next step

No

Install the breather tube and route it correctly
Attempt 10 starts to the vehicle. If the vehicle does **NOT** start during any of the 10 attempts. Install new starter motor
If the vehicle **DOES** start during all of the 10 attempts
Proceed to next step

A5: STARTER MOTOR CHECKS



NOTE:

This test will require the technician to listen for audible result, high surrounding noise may hinder the outcome

1 Set the ignition ON

2 Depress the brake pedal

3 Depress the start button

Was there an audible click when the start button was pressed and did the vehicle start?
(Audible click will not be heard if the vehicle starts)

Yes

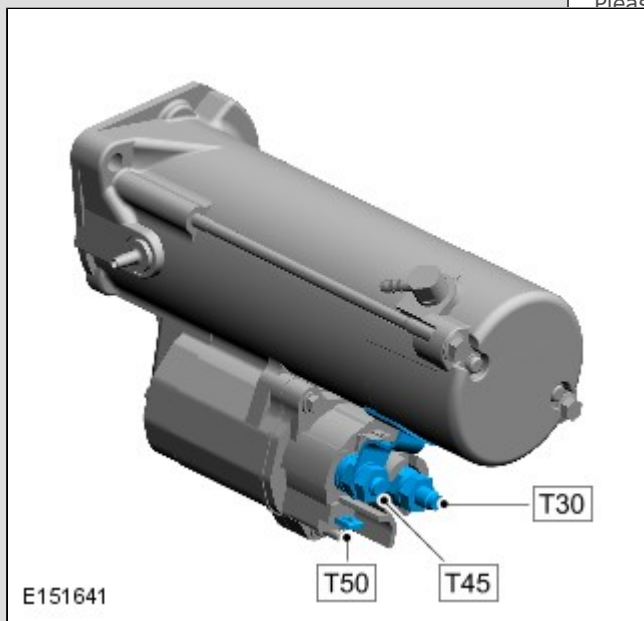
Proceed to next step

No

Set the ignition OFF. Repeat this starter motor check a further 10 times. Record the result for number of none click tests
If there was an audible click when the start button was pressed during the 10 test cycle. Proceed to next step

A6: CHECK CONNECTIONS ON THE STARTER MOTOR

Please refer to workshop manual section 303-
Specification section



2 Gain access to the starter motor

3 Refer to the electrical circuit diagrams and check the battery power supply and terminal to the starter motor (battery positive cable nut T30 permanent feed B1)

4 Refer to the electrical circuit diagrams and check the operating supply and terminal to the starter motor (switch lead to the starter motor solenoid nut T50)

5 Refer to the electrical circuit diagrams and check the power feed stud on the starter motor (T45 terminal)

6 Refer to the electrical circuit diagrams and inspect connections for signs of becoming loose, water ingress, and terminals/stud for damage and/or corrosion

Are terminal/stud connections to starter motor, loose, damaged or corroded?

Yes

Rectify as required
Re-check the system is operating correctly
Return vehicle to the customer

No

Proceed to next step

A7: VOLTAGE DROP GROUND CIRCUIT



WARNING:

During this test be aware of hot or moving parts. Failure to follow this instruction may result in personal injury

**NOTES:**

- You will require assistance during this test
- You may require extended test leads for the multimeter

1 Connect the multimeter between the battery negative terminal and the starter motor ground (where bolted to the bell housing)

2 Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading

Is reading equal to or below 0.6 volts?

Yes

Proceed to next step

No

Refer to the electrical circuit diagrams and check ground circuit between battery negative terminal and starter motor for corrosion and high resistance. Rectify as required. Re-check the system is operating correctly. Return vehicle to the customer

A8: VOLTAGE DROP POWER CIRCUIT**WARNING:**

During this test be aware of hot or moving parts. Failure to follow this instruction may result in personal injury

**NOTES:**

- You will require assistance during this test
- You may require extended test leads for the multimeter

1 Connect the multimeter between the battery positive terminal and the power feed stud on the starter motor (T45 terminal)

2 Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading

Is reading equal to or below 0.6 volts?

Yes

Proceed to next step

No

Refer to the electrical circuit diagrams and check power circuit between battery positive terminal and the power feed stud on the starter motor (T45 terminal) including mega fuses, for corrosion and high resistance. Rectify as

required
Re-check the system is operating correctly
Return vehicle to the customer

A9: VOLTAGE AT STARTER MOTOR (T50)



WARNING:

During this test be aware of hot or moving parts. Failure to follow this instruction may result in personal injury



NOTES:

- You will require assistance during this test
- You may require extended test leads for the multimeter

1 Check the operating supply voltage to the starter motor (T50)

2 Connect the multimeter between the starter motor (T50) terminal and ground

3 Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading

Is the reading between 9 & 12 volts during cranking?

Yes

Proceed to next step

No

Refer to the electrical circuit diagrams and check starter relay circuit for open circuit, high resistance
Rectify as required
Re-check the system is operating correctly
Return vehicle to the customer

A10: VOLTAGE AT STARTER MOTOR SOLENOID (T45)



WARNING:

During this test be aware of hot or moving parts. Failure to follow this instruction may result in personal injury



NOTE:

You will require assistance during this test

1 Check the operating supply voltage to the starter motor (T45)

	2 Connect the multimeter between the starter motor (T45) terminal and ground
	3 Set the multimeter to read DC voltage. Set the ignition ON. Depress the brake pedal. Depress the start button and record the reading
	Is the reading between 6 & 9 volts during cranking? Yes Proceed to next step No Check and install new starter motor and breather tube

A11: STARTER MOTOR VISUAL INSPECTION



WARNING:

During this test isolate the vehicle battery. Failure to follow this instruction may result in personal injury

	1 Remove the starter motor
	2 Visually inspect the starter motor ring gear for failure
	3 Visually inspect the flywheel ring gear for failure
	Was there evidence of failure to the starter motor or flywheel ring gear Yes Check and install new parts as requiredRe-check the system is operating correctlyReturn vehicle to the customer No Contact dealer technical support with all of the recorded values from the above tests

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4L Diesel, DTC: Module Name: Engine Control Module (100-00 General Information, Description and Operation).